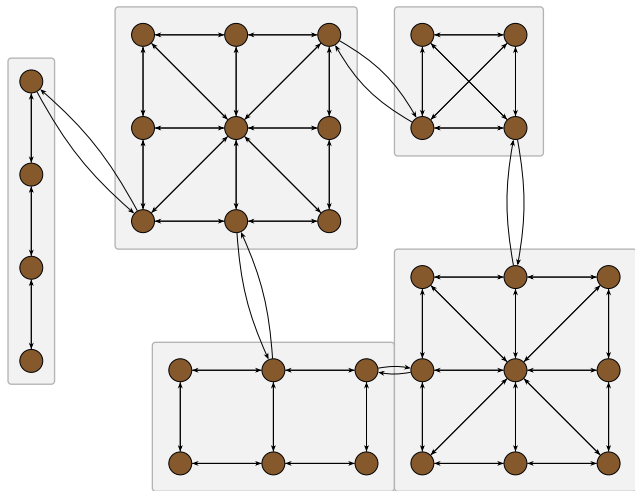


3. Markov models

- Markov chains
 - Inference
 - Parameter learning
 - Convergence
- Hidden Markov models
 - Inference
 - Decoding
 - Parameter learning
 - Continuous observation space



Outline

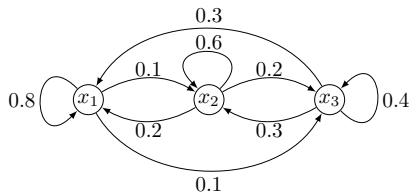
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Markov chains

Markov chain: a stochastic process $\{X_0, X_1, \dots, X_t, \dots\}$ with Markov property

$$\mathbf{P}(x_{t+1} \mid x_0, \dots, x_t) = \mathbf{P}(x_{t+1} \mid x_t)$$

- X : state space
- time-homogeneous if $\mathbf{P}(X_{t+1} = x_j \mid X_t = x_i)$ is constant for all t



Markov chains

initial state distribution

$$\rho = (\dots, \mathbf{P}(X_0 = x_i), \dots), \quad i = 1, \dots, m$$

- $\rho \in \mathbf{R}^m$; $\rho \succeq 0$; $\mathbf{1}^T \rho = 1$

transition matrix

$$P_{ij} = \mathbf{P}(x_j \mid x_i), \quad \text{for all } x_i, x_j \in X$$

- $P \in \mathbf{R}^{m \times m}$
- $P_{ij} \geq 0$ for all $i = 1, \dots, m, j = 1, \dots, m$
- $\mathbf{1}^T P_{i:} = 1$ for all $i = 1, \dots, m$

Inference

given:

- parameters $\Theta = \{\rho, P\}$ of a Markov chain
- observed sequence of states $\zeta = \{X_0 = x_i, X_1 = x_j, X_2 = x_k, \dots\}$

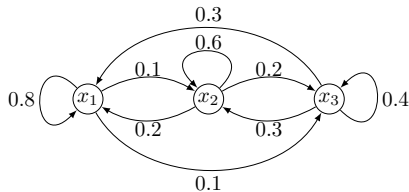
calculate the probability of observing the sequence ζ

$$\mathbf{P}(\zeta \mid \rho, P) = \rho_i P_{ij} P_{jk} \cdots$$

example

$$\rho = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right) \quad P = \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.2 & 0.6 & 0.2 \\ 0.3 & 0.3 & 0.4 \end{bmatrix}$$

$$\mathbf{P}(x_1, x_2, x_3, x_1 \mid \rho, P) = \frac{1}{3} \times 0.1 \times 0.2 \times 0.3$$



Parameter learning

given a set of observed state sequences $\mathcal{D} = \{\zeta_1, \zeta_2, \dots\}$ with $\zeta = \{x_0, \dots, x_N\}$

estimate $\Theta = \{\rho, P\}$

- learning initial state distribution ρ :

$$\rho_i = \mathbf{E}_{\zeta \sim \mathcal{D}} \mathbf{P}(X_0 = x_i \mid \zeta), \quad i = 1, \dots, m$$

- learning transition matrix P :

$$P_{ij} = \frac{\mathbf{E}_{\zeta \sim \mathcal{D}, t} \mathbf{P}(X_t = x_i, X_{t+1} = x_j \mid \zeta)}{\mathbf{E}_{\zeta \sim \mathcal{D}, t} \mathbf{P}(X_t = x_i \mid \zeta)}, \quad i = 1, \dots, m, \quad j = 1, \dots, m$$

$$- \quad t = 0, \dots, N - 1$$

Parameter learning

optimization problem

$$\begin{aligned} &\text{maximize} && l_{\mathcal{D}}(\Theta) = \mathbf{E}_{\zeta \sim \mathcal{D}} \log \mathbf{P}(\zeta \mid \rho, P) \\ &\text{subject to} && \rho \succeq 0, \quad \mathbf{1}^T \rho = 1 \\ &&& P_{ij} \geq 0, \quad \mathbf{1}^T P_{i:} = 1, \quad i = 1, \dots, m, \quad j = 1, \dots, m \end{aligned}$$

$$\begin{aligned} l_{\mathcal{D}}(\Theta) &= \mathbf{E}_{\zeta \sim \mathcal{D}} \left(\log \mathbf{P}(x_0 \mid \rho) \prod_{t=0}^{N-1} \mathbf{P}(x_{t+1} \mid x_t, P) \right) \\ &= \mathbf{E}_{\zeta \sim \mathcal{D}} \log \mathbf{P}(x_0 \mid \rho) + \mathbf{E}_{\zeta \sim \mathcal{D}} \left(\sum_{t=0}^{N-1} \log \mathbf{P}(x_{t+1} \mid x_t, P) \right) \\ &= \mathbf{E}_{\zeta \sim \mathcal{D}} \left(\sum_{i=1}^m I_{x_i}(x_0) \log \rho_i \right) + \mathbf{E}_{\zeta \sim \mathcal{D}} \left(\sum_{i=1}^m \sum_{j=1}^m \sum_{t=0}^{N-1} I_{x_i}(x_t) I_{x_j}(x_{t+1}) \log P_{ij} \right) \end{aligned}$$

Parameter learning

example

observed state sequences from a 3-states Markov chain:

- $(x_2, x_2, x_3, x_3, x_3, x_3, x_1)$
- $(x_1, x_3, x_2, x_3, x_3, x_3, x_3)$
- (x_3, x_3, x_2, x_2)
- $(x_2, x_1, x_2, x_2, x_1, x_3, x_1)$

$$\begin{aligned}\rho &= \left(\frac{1}{4}, \frac{2}{4}, \frac{1}{4}\right) \\ &= (0.25, 0.5, 0.25)\end{aligned}$$

$$P = \begin{bmatrix} \frac{0}{3} & \frac{1}{3} & \frac{2}{3} \\ \frac{2}{7} & \frac{3}{7} & \frac{2}{7} \\ \frac{2}{11} & \frac{2}{11} & \frac{7}{11} \end{bmatrix}$$

Convergence

requirements

- irreducible: $P_{ij}^t > 0$ for some t and all $x_i, x_j \in X$
- aperiodic: no fixed interval returning back to the same state

convergence: when $t \rightarrow \infty$

$$P^T \pi = \pi$$

$$\lim_{t \rightarrow \infty} P^t = \mathbf{1} \pi^T$$

- $\pi \in \mathbf{R}_+^m$: stationary state distribution

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Hidden markov models

hidden Markov model (HMM) consists of

- a Markov chain $\{X_0, X_1, \dots, X_t, \dots\}$ with states not observable
- an observable stochastic process $\{Y_0, Y_1, \dots, Y_t, \dots\}$ with

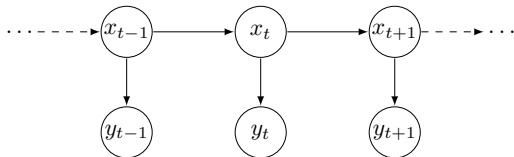
$$\mathbf{P}(y_t \mid x_0, \dots, x_t) = \mathbf{P}(y_t \mid x_t)$$

– Y : observation space

emission matrix

$$B_{ij} = \mathbf{P}(y_j \mid x_i)$$

- $B \in \mathbf{R}^{m \times n}$
- $B_{ij} \geq 0$ for all $i = 1, \dots, m, j = 1, \dots, n$
- $\mathbf{1}^T B_{i:} = 1$ for all $i = 1, \dots, m$



Inference

given:

- parameters $\Theta = \{\rho, P, B\}$ of an HMM
- some sequence of observations $\varphi = \{y_0, \dots, y_N\}$

calculate the probability of observing the observation sequence φ

direct method

$$\begin{aligned}\mathbf{P}(\varphi \mid \Theta) &= \sum_{\zeta} \mathbf{P}(\varphi, \zeta \mid \Theta) \\ &= \sum_{x_0, \dots, x_N} \mathbf{P}(x_0 \mid \rho) \mathbf{P}(y_0 \mid x_0, B) \prod_{t=0}^{N-1} \mathbf{P}(x_{t+1} \mid x_t, P) \mathbf{P}(y_{t+1} \mid x_{t+1}, B)\end{aligned}$$

- complexity: $2Nm^N$ (or simply m^N)

Inference

forward algorithm

$$\alpha_t(i) = \mathbf{P}(y_0, \dots, y_t, X_t = x_i \mid \Theta), \quad i = 1, \dots, m$$

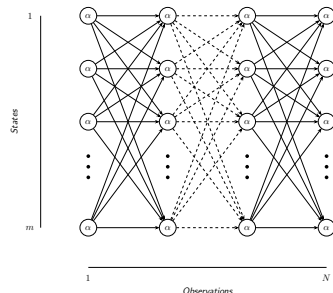
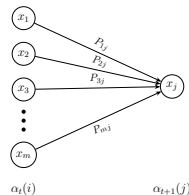
- $\alpha_t \in \mathbf{R}_+^m$: forward probability
- recursive expression:

$$\alpha_t(i) = \begin{cases} \rho_i \mathbf{P}(y_0 \mid X_0 = x_i, B), & t = 0 \\ \alpha_{t-1}^T P_{:,i} \mathbf{P}(y_t \mid X_t = x_i, B), & t > 0 \end{cases}$$

for $i = 1, \dots, m$

- probability of observing the observation sequence φ

$$\mathbf{P}(\varphi \mid \Theta) = \mathbf{1}^T \alpha_N$$



Inference

forward algorithm

given hidden Markov model parameters Θ , observation sequence φ .

for $i = 1, \dots, m$:

$$\alpha_0(i) := \rho_i \mathbf{P}(y_0 \mid X_0 = x_i, B).$$

for $t = 1, \dots, N$, $i = 1, \dots, m$:

$$\alpha_t(i) := \alpha_{t-1}^T P_{:,i} \mathbf{P}(y_t \mid X_t = x_i, B).$$

return $\mathbf{P}(\varphi \mid \Theta) = \mathbf{1}^T \alpha_N$.

- complexity: Nm^2 (or simply m^2)

Decoding

given:

- parameters $\Theta = \{\rho, P, B\}$ of an HMM
- some sequence of observations $\varphi = \{y_0, \dots, y_N\}$

find:

- the most probable state x_t^* at time t
- the most probable state sequence $\zeta^* = \{x_0^*, \dots, x_N^*\}$ that generated φ

Optimal state prediction

$$\beta_t(i) = \mathbf{P}(y_{t+1}, \dots, y_N \mid X_t = x_i, \Theta), \quad i = 1, \dots, m$$

- $\beta_t \in \mathbf{R}_+^m$: backward probability
- recursive expression:

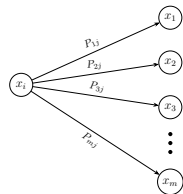
$$\beta_t(i) = \begin{cases} \sum_{j=1}^m \beta_{t+1}(j) P_{ij} \mathbf{P}(y_{t+1} \mid X_{t+1} = x_j, B), & t < N \\ 1, & t = N \end{cases}$$

for $i = 1, \dots, m$

- probability of observing the observation sequence φ

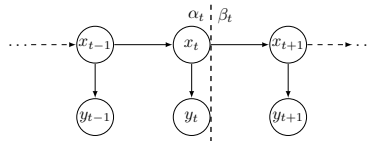
$$\mathbf{P}(\varphi \mid \Theta) = \alpha_t^T \beta_t$$

for all $t = 1, \dots, N$



$\beta_t(i)$

$\beta_{t+1}(j)$



Optimal state prediction

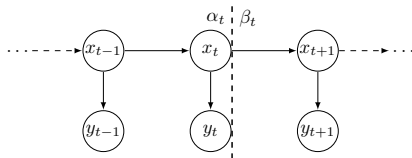
$$\gamma_t(i) = \mathbf{P}(X_t = x_i \mid \varphi, \Theta), \quad i = 1, \dots, m$$

- $\gamma_t \in \mathbf{R}_+^m$
- in terms of α_t and β_t :

$$\gamma_t(i) = \frac{\mathbf{P}(X_t = x_i, \varphi \mid \Theta)}{\mathbf{P}(\varphi \mid \Theta)} = \frac{\alpha_t(i)\beta_t(i)}{\alpha_t^T \beta_t}, \quad i = 1, \dots, m$$

- the most probable state x_t^* given φ and Θ

$$x_t^* = \operatorname{argmax}_{x_i} \gamma_t(i)$$



Optimal sequence prediction

$$x_0^*, \dots, x_N^* = \operatorname{argmax}_{x_0, \dots, x_N} \mathbf{P}(x_0, \dots, x_N \mid \varphi, \Theta)$$

approximate solution

$$\tilde{\zeta}^* = \left\{ \operatorname{argmax}_{x_i} \gamma_t(i) \mid t = 0, \dots, N \right\}$$

- computationally efficient
- not the global optimal
- does not consider state transitions

Optimal sequence prediction

$$\mathbf{P}(\zeta \mid \varphi, \Theta) = \frac{\mathbf{P}(\zeta, \varphi \mid \Theta)}{\mathbf{P}(\varphi \mid \Theta)} \propto \mathbf{P}(\zeta, \varphi \mid \Theta)$$

Viterbi algorithm

$$\begin{aligned}\delta_t(i) &= \max_{x_0, \dots, x_{t-1}} \mathbf{P}(x_0, \dots, x_{t-1}, X_t = x_i, y_0, \dots, y_t \mid \Theta) \\ &= \max_{j=1, \dots, m} (\delta_{t-1}(j) P_{ji}) \mathbf{P}(y_t \mid X_t = x_i, B), \quad i = 1, \dots, m\end{aligned}$$

- $\delta_t \in \mathbf{R}_+^m$: probability of being in state x_i at time t given that the state subsequence up until $t - 1$ is optimal w.r.t. the partial observation sequence $\{y_0, \dots, y_{t-1}\}$

$$\psi_t(i) = \operatorname{argmax}_{j=1, \dots, m} \delta_{t-1}(j) P_{ji}, \quad i = 1, \dots, m$$

- $\psi_t \in \mathbf{Z}_{++}^m$: the index j of the previous state x_j at time $t - 1$ that gives the maximum probability $\delta_{t-1}(j) P_{ji}$, for each state i at time t

Optimal sequence prediction

Viterbi algorithm

given hidden Markov model parameters Θ , observation sequence φ .

1. *Initialization.* **for** $i = 1, \dots, m$:

$$\delta_0(i) := \rho_i \mathbf{P}(y_0 \mid X_0 = x_i, B).$$

2. *Recursion.* **for** $t = 1, \dots, N$, $i = 1, \dots, m$:

$$\delta_t(i) := \max_{j=1, \dots, m} (\delta_{t-1}(j) P_{ji}) \mathbf{P}(y_t \mid X_t = x_i, B).$$

$$\psi_t(i) := \operatorname{argmax}_{j=1, \dots, m} \delta_{t-1}(j) P_{ji}.$$

3. *Termination.*

$$p^* := \max_{i=1, \dots, m} \delta_N(i).$$

$$i_N^* := \operatorname{argmax}_{i=1, \dots, m} \delta_N(i).$$

$$x_N^* := x_{i_N^*}.$$

4. *Backtracking.* **for** $t = N, \dots, 1$:

$$i_{t-1}^* := \psi_t(i_t^*).$$

$$x_{t-1}^* := x_{i_{t-1}^*}.$$

Parameter learning

expectation-maximization (EM) algorithm

-
1. initialize parameters Θ
 2. E-step: calculate some likelihood function w.r.t. Θ
 3. M-step: update Θ by maximizing the likelihood function from 2
 4. repeat 2–3 until convergence
-

- can be proved to converge to local optimum
- not guarantee to find the global optimum

Parameter learning

for each EM-iteration

- $\Theta = \{\rho, P, B\}$: the set of estimated parameters from previous iteration
- $\Theta^+ = \{\rho^+, P^+, B^+\}$: the new set of parameters to be updated

$$\rho_i^+ = \mathbf{E}_{\varphi \sim \mathcal{D}} \mathbf{P}(X_0 = x_i \mid \varphi, \Theta), \quad i = 1, \dots, m$$

$$P_{ij}^+ = \frac{\mathbf{E}_{\varphi \sim \mathcal{D}, t} \mathbf{P}(X_t = x_i, X_{t+1} = x_j \mid \varphi, \Theta)}{\mathbf{E}_{\varphi \sim \mathcal{D}, t} \mathbf{P}(X_t = x_i \mid \varphi, \Theta)}, \quad i = 1, \dots, m, \quad j = 1, \dots, m$$

$$B_{ij}^+ = \frac{\mathbf{E}_{\varphi \sim \mathcal{D}, t} \mathbf{P}(X_t = x_i, Y_t = y_j \mid \varphi, \Theta)}{\mathbf{E}_{\varphi \sim \mathcal{D}, t} \mathbf{P}(X_t = x_i \mid \varphi, \Theta)}, \quad i = 1, \dots, m, \quad j = 1, \dots, n$$

- $t = 0, \dots, N - 1$

Parameter learning

Baum-Welch algorithm

$$\begin{aligned}\xi_t(i, j) &= \mathbf{P}(X_t = x_i, X_{t+1} = x_j \mid \varphi, \Theta) \\ &= \frac{\mathbf{P}(X_t = x_i, X_{t+1} = x_j, \varphi \mid \Theta)}{\mathbf{P}(\varphi \mid \Theta)}, \quad i = 1, \dots, m, \quad j = 1, \dots, m\end{aligned}$$

- $\xi_t \in \mathbf{R}_+^{m \times m}$: probability of transitioning from state x_i at t to state x_j at $t + 1$ given φ
- in terms of α_t and β_t :

$$\xi_t(i, j) = \frac{\alpha_t(i)P(i, j)\mathbf{P}(y_{t+1} \mid X_{t+1} = x_j, B)\beta_{t+1}(j)}{\sum_{i=1}^m \sum_{j=1}^m \alpha_t(i)P(i, j)\mathbf{P}(y_{t+1} \mid X_{t+1} = x_j, B)\beta_{t+1}(j)}$$

for all $i = 1, \dots, m, j = 1, \dots, m$

- $\gamma_t(i) = \sum_{j=1}^m \xi_t(i, j)$, for all $i = 1, \dots, m$

Parameter learning

Baum-Welch algorithm

$$\rho_i^+ = \mathbf{E}_{\varphi \sim \mathcal{D}} \gamma_0(i), \quad i = 1, \dots, m$$

$$P_{ij}^+ = \frac{\mathbf{E}_{\varphi \sim \mathcal{D}, t} \xi_t(i, j)}{\mathbf{E}_{\varphi \sim \mathcal{D}, t} \gamma_t(i)}, \quad i = 1, \dots, m, \quad j = 1, \dots, m$$

$$B_{ij}^+ = \frac{\mathbf{E}_{\varphi \sim \mathcal{D}, t} (\gamma_t(i) I_j(y_t))}{\mathbf{E}_{\varphi \sim \mathcal{D}, t} \gamma_t(i)}, \quad i = 1, \dots, m, \quad j = 1, \dots, n$$

- indicator function $I_j(y) = 1$ if the index of $y \in Y$ is equal to j , and 0 otherwise

Continuous observation space

Gaussian hidden Markov models

- $\text{dom } Y = \mathbf{R}$
- the emission map from X to Y is given by the Gaussian density function $\mathcal{N}(\mu_i, \sigma_i^2)$:

$$p(y \mid X = x_i) = \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left(-\frac{(y - \mu_i)^2}{2\sigma_i^2}\right), \quad i = 1, \dots, m$$

- $\mu_i \in \mathbf{R}$: mean of the Gaussian distribution
- $\sigma_i^2 \in \mathbf{R}$: variance of the Gaussian distribution